

PPO: Proximal Policy Optimisation

The clipped surrogate objective that prevents catastrophic policy collapse — the engine of RLHF Stage 3

10 min

Duration

B3 – Apply

Bloom's

L3 – Advanced

Level

AI-RL-IN-TH-000004

ID

PPO Clipped Surrogate Objective — Full Derivation

Problem with naive policy gradient: large steps → policy collapse

Standard PG: $L^{PG}(\theta) = E[r_t(\theta) \cdot A_t]$

where $r_t(\theta) = \pi_\theta(a_t|s_t) / \pi_{old}(a_t|s_t)$ (probability ratio)

and A_t = advantage estimate (how much better than expected)

PPO solution: clip the probability ratio to $[1-\epsilon, 1+\epsilon]$

$L^{CLIP}(\theta) = E[\min(r_t(\theta) \cdot A_t, \text{clip}(r_t(\theta), 1-\epsilon, 1+\epsilon) \cdot A_t)]$

Typical $\epsilon = 0.1$ to 0.2 . The clip prevents $r_t(\theta)$ from exceeding $1+\epsilon$ or going below $1-\epsilon$.

Effect: if the new policy moves too far from the old policy, the gradient is cut off

PPO in the LLM Context — Unique Challenges

Episode Length

One 'episode' = one full response generation (often 200+ tokens). The episode length is orders of magnitude longer than typical RL tasks (games, robotics).

Sparse Reward

The reward model provides only one score per full response — at the end-of-sequence token. All intermediate tokens are trained on zero reward (only KL penalty).

Actor-Critic Setup

Actor (policy): generates text. Critic (value model): estimates expected future reward from current state. Both require full-model-scale networks.

4 Model Copies

Actor + Critic + Reference + Reward Model. Each is a copy of the base LLM. Total VRAM = $4 \times$ model size. Requires large GPU clusters.

PPO Training Loop — Pseudocode Walkthrough

Outer Loop (per batch)

- 1. Sample prompt x from dataset
- 2. Generate response $y \sim \pi_{\theta}(\cdot | x)$ (actor)
- 3. Score: $r = \text{reward_model}(x, y)$
- 4. Compute KL: $kl = \text{KL}(\pi_{\theta} \parallel \pi_{\text{SFT}})$ per token
- 5. Total reward = $r - \beta \cdot kl$

Inner Loop (PPO update)

- 6. Compute advantage A_t with critic V_{ϕ}
- 7. Compute $r_t(\theta) = \pi_{\theta}(a_t) / \pi_{\text{old}}(a_t)$
- 8. Compute $L^{\text{CLIP}}(\theta)$ for actor
- 9. Compute value loss for critic
- 10. Update both actor and critic

KEY REFERENCES

- Schulman, J. et al. (2017). Proximal Policy Optimisation Algorithms. arXiv:1707.06347
- Ouyang, L. et al. (2022). InstructGPT. NeurIPS 2022.
- TRL PPOTrainer. huggingface.co/docs/trl/ppo_trainer

 **Next: Concept 23: Goodhart's Law & DPO (AI-RL-IN-TH-000005)**